

AWS → CLOUDFLARE MIGRATION · PROJECT DOCUMENTATION

Migration & Cutover Runbook

Supporting Document 3 - Step-by-Step Cutover, Validation & Rollback

SaaS Edge Layer - AWS CloudFront/WAF/Shield → Cloudflare

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Audience: Client / CTO / Engineering Team

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Introduction

This runbook is the step-by-step guide used to migrate the public edge to Cloudflare safely, with a rollback path available at every stage. It is written so the steps can be followed in order, and so the client can understand exactly what was done and how to reverse it if needed. Checkbox items can be ticked off as each step is completed.

Golden rule

Nothing touches production until it has been verified on a sandbox, and the old AWS edge stays in place and billable until the new setup is confirmed working. Decommissioning happens last.

1. Pre-Migration Checklist

Complete these before making any changes.

- Confirm full inventory of current DNS records in Route 53.
- Export the complete list of AWS WAF rules (managed, custom, rate, IP, geo, bot).
- Document current CloudFront cache behaviors and origins.
- Record current SSL/TLS certificates and how they are issued.
- Note the Shield Advanced subscription terms and renewal/billing window.
- Identify the ALB DNS name that will become the Cloudflare origin.
- Confirm a sandbox hostname is available for testing.
- Agree a cutover window and rollback decision points with the client.

2. Cloudflare Account & Zone Setup

1. Create the Cloudflare account and select the Business plan.
2. Add the client's domain as a new zone.
3. Note the Cloudflare-assigned nameservers (used later at cutover).
4. Set the zone to the correct configuration before importing records.

Do not change nameservers yet

Adding the zone does not move traffic. The domain only moves to Cloudflare once the nameservers are switched in Section 9 - after testing is complete.

3. DNS Import & Configuration

Recreate the DNS records in Cloudflare to match Route 53 exactly.

1. Import all DNS records into the Cloudflare zone.
2. Verify each record matches Route 53 (type, name, value, TTL).
3. Set the main application records to proxied (orange-cloud) so traffic flows through Cloudflare.
4. Leave records that should bypass Cloudflare as DNS-only where appropriate.
5. Point the proxied origin at the AWS ALB DNS name.

4. SSL/TLS Configuration - Full Strict Mode

Configure SSL/TLS so traffic is encrypted end-to-end, with certificates validated all the way to the origin.

1. Set the SSL/TLS mode to Full (Strict).
2. Ensure the AWS ALB presents a valid certificate that Cloudflare will trust.
3. Confirm the edge certificate is active for the domain.
4. Verify HTTPS works through Cloudflare to the origin on the sandbox host.

SSL/TLS mode	Meaning	Used here?
Flexible	Encrypts user→Cloudflare only	No
Full	Encrypts end-to-end, cert not strictly validated	No
Full (Strict)	Encrypts end-to-end, origin cert validated	Yes

5. CDN Cache Rules

Recreate the caching behavior CloudFront provided, using Cloudflare cache rules.

1. Define which paths and file types should be cached at the edge.
2. Set cache TTLs to match the previous CloudFront behavior.
3. Ensure dynamic and authenticated routes are not cached.
4. Verify cache hits and misses behave correctly on the sandbox.

6. WAF & Firewall Rules

Apply the full ruleset from the WAF Rule Mapping document (Supporting Document 2).

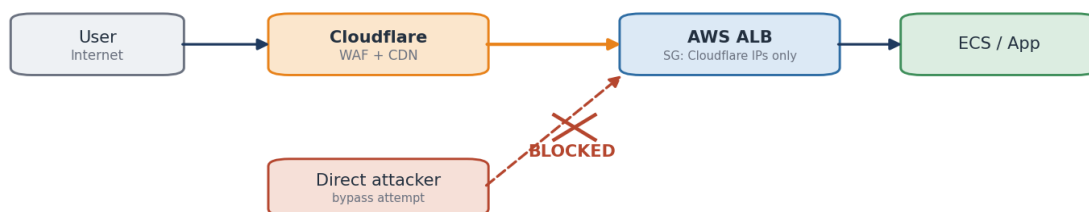
- Enable the Cloudflare managed rulesets.
- Recreate all custom firewall rules.
- Configure rate limiting rules to match AWS thresholds.
- Import IP allow and block lists.
- Configure country-based rules.
- Enable bot protection.

See *Supporting Document 2* for the full rule-by-rule mapping and validation checklist.

7. Origin Protection - ALB Security Group to Cloudflare IPs

This is the step many migrations miss. Without it, an attacker could bypass Cloudflare and hit the AWS origin directly, defeating the whole edge layer.

Origin Protection — Locking the ALB to Cloudflare



The ALB security group only accepts Cloudflare IP ranges, so no one can bypass Cloudflare and hit the origin directly.

Figure 1 - ALB locked to Cloudflare IP ranges

1. Obtain the current Cloudflare IP ranges.
2. Update the ALB security group to allow inbound traffic only from those ranges.
3. Remove any rules that allow direct public access to the ALB.
4. Confirm the application still works through Cloudflare after the lockdown.

Timing

Apply the origin lock as part of cutover, once traffic is flowing through Cloudflare not before, or legitimate traffic still on the AWS edge could be blocked.

8. Staging Test Procedure

Verify everything on the sandbox before any real traffic moves.

- DNS resolves correctly through Cloudflare for the sandbox host.
- HTTPS works end-to-end in Full Strict mode.
- Cache rules produce the expected hits and misses.
- WAF rules block what they should and allow legitimate traffic.
- Rate limiting triggers at the configured thresholds.
- Traffic routes correctly to the AWS ALB origin.

Gate

Do not proceed to production cutover until every item in this section passes. This is the main safety gate of the migration.

9. Production Cutover Steps

With sandbox testing clean, move production to Cloudflare.

1. Lower DNS TTLs ahead of time so changes propagate quickly.
2. Change the domain's nameservers to the Cloudflare nameservers.
3. Confirm Cloudflare is now authoritative and proxying production traffic.
4. Apply the ALB origin lock (Section 7) so only Cloudflare can reach the origin.
5. Watch traffic and error rates closely during propagation.

10. Post-Cutover Validation Checklist

- Real user traffic is arriving through Cloudflare.
- ALB access logs show traffic only from Cloudflare IP ranges.
- Application error rates are normal.
- WAF events show rules firing as expected.
- Cache performance is as expected in Cloudflare analytics.
- SSL/TLS is valid and Full Strict is active.

Monitor these for about a week after cutover to confirm real traffic is unaffected.

11. Rollback Procedure

If a serious problem appears during or shortly after cutover, traffic can be returned to the AWS edge because it is still in place and was not decommissioned until after validation.

1. Revert the domain nameservers back to the previous Route 53 configuration.
2. If the origin lock was applied, restore the ALB security group to its prior state.
3. Confirm traffic is flowing again through the original AWS edge.
4. Investigate the issue on the sandbox before attempting cutover again.

Why rollback is low-risk here

Because the AWS edge stack is left running and billable until after validation, rolling back is simply a matter of pointing DNS back. No AWS edge service is removed until the new setup is proven.

12. Shield Advanced Cancellation Process

Shield Advanced carries a subscription commitment, so it is handled carefully and last, never removed blindly. Only after the Cloudflare setup is fully validated is the AWS edge decommissioned.

1. Confirm Cloudflare has been validated and stable in production.
2. Review the Shield Advanced subscription term and renewal/billing window.
3. Schedule the unsubscribe to align with the renewal window, avoiding penalties.
4. Unsubscribe Shield Advanced.
5. Decommission Firewall Manager.
6. Decommission CloudFront.
7. Confirm the final edge bill reflects only the Cloudflare cost.

End state

Once decommissioning is complete, the public edge runs entirely on Cloudflare at around \$200/month, the AWS application stack is unchanged, and the origin is locked to Cloudflare — delivering the cost saving safely and without disruption.